

Correlation

In a bivariate distribution, if the change in one variable affects a change in the other variable, the variables are said to be correlated.

Types of correlation

(i) Positive correlation: If the values of the two variables deviate in the same direction i.e., if the increase (or decrease) in one results in a corresponding increase (or decrease) in the other, correlation is said to be direct or positive.

Example of positive correlation are -

- The family income and expenditure on luxury items.
- Price and supply of a commodity.

(ii) Negative correlation: If the two variables deviate in opposite direction i.e., if the increase (or decrease) in one results in a corresponding decrease (or increase) in the other, correlation is said to be inverse or negative.

Example of negative correlation are -

- Price and demand of a commodity.
- Volume and pressure of a perfect gas.

Reasons Responsible for the existence of correlation :-

1. Due to the presence of cause-effect relationship between variables : For example, a high degree correlation between 'temperature' and 'sale of coffee' is due to the fact that people like taking coffee in winter season.
2. Due to the presence of interdependent relationship between the variables : for example, the presence of correlation between amount spent on entertainment of family and total expenditure of family is due to fact that both variables affect each other.

3. Due to effect of some common cause:

For example, positive correlation between the numbers of girls seeking admission in colleges A and B of a city may be due to the effect of increasing interest of girls towards higher education.

Karl Pearson's co-efficient of correlation
(or Product moment correlation coefficient)

Correlation co-efficient between two variables x and y , usually denoted by $r(x, y)$ or r_{xy} is a numerical measure of linear relationship between them and is defined as

$$\begin{aligned} r_{xy} &= \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}} \\ &= \frac{\frac{1}{n} \sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\frac{1}{n} \sum (x_i - \bar{x})^2 \cdot \frac{1}{n} \sum (y_i - \bar{y})^2}} = \frac{\frac{1}{n} \sum (x_i - \bar{x})(y_i - \bar{y})}{\sigma_x \sigma_y} \end{aligned}$$

$\therefore$

$$r_{xy} = \frac{\sum (x - \bar{x})(y - \bar{y})}{n \sigma_x \sigma_y}$$

Alternate form of $r(x, y)$!

$$r(x, y) = \frac{n \sum xy - \sum x \sum y}{\sqrt{n \sum x^2 - (\sum x)^2} \sqrt{n \sum y^2 - (\sum y)^2}}$$

Ex.1 find the coefficient of correlation between the values of x and y :

x	1	3	5	7	8	10
y	8	12	15	17	18	20

Solⁿ Here, $n = 6$

x	y	x^2	y^2	xy
1	8	1	64	8
3	12	9	144	36
5	15	25	225	75
7	17	49	289	119
8	18	64	324	144
10	20	100	400	200
$\sum x = 34$	$\sum y = 90$	$\sum x^2 = 248$	$\sum y^2 = 1446$	$\sum xy = 582$

Karl Pearson's coefficient of correlation is given by

$$r(x, y) = \frac{n \sum xy - \sum x \sum y}{\sqrt{n \sum x^2 - (\sum x)^2} \sqrt{n \sum y^2 - (\sum y)^2}}$$

$$= \frac{(6 \times 582) - (34 \times 90)}{\sqrt{(6 \times 248) - (34)^2} \sqrt{(6 \times 1446) - (90)^2}}$$

$$= 0.9879$$

Ex. 2 The following data regarding the heights (y) and weight (x) of 100 college students are given:

$$\sum x = 15000, \sum x^2 = 2272500, \sum y = 6800$$

$$\sum y^2 = 463025, \sum xy = 1022250$$

Find the correlation coefficient between height and weight.

Solⁿ Here, $n = 100$

Correlation coefficient $r(x, y)$ is given by

$$r(x, y) = \frac{n \sum xy - \sum x \sum y}{\sqrt{n \sum x^2 - (\sum x)^2} \sqrt{n \sum y^2 - (\sum y)^2}}$$

$$= \frac{(100 \times 1022250) - (15000 \times 6800)}{\sqrt{(100 \times 2272500) - (15000)^2} \sqrt{(100 \times 463025) - (6800)^2}}$$

$$= 0.6$$

Ex.3 Calculate the coefficient of correlation for the ages of husbands and wives:

Ages of husband (Years)	23	27	28	29	30	31	33	35	36	39
Ages of wife (Year)	18	22	23	24	25	26	28	29	30	32

Solⁿ calculation for correlation coefficient

x	y	$u = x - 31$	$v = y - 25$	u^2	v^2	uv
23	18	-8	-7	64	49	56
27	22	-4	-3	16	9	12
28	23	-3	-2	9	4	6
29	24	-2	-1	4	1	2
30	25	-1	0	1	0	0
31	26	0	1	0	1	0
33	28	2	3	4	9	6
35	29	4	4	16	16	16
36	30	5	5	25	25	25
39	32	8	7	64	49	56
$\Sigma x = 311$	$\Sigma y = 257$	$\Sigma u = 1$	$\Sigma v = 7$	$\Sigma u^2 = 203$	$\Sigma v^2 = 163$	$\Sigma uv = 179$

Karl Pearson coefficient between u & v is given by

$$r_{uv} = \frac{n \sum uv - (\sum u)(\sum v)}{\sqrt{n \sum u^2 - (\sum u)^2} \sqrt{n \sum v^2 - (\sum v)^2}}$$

$$= \frac{(10 \times 179) - (1 \times 7)}{\sqrt{(10 \times 203) - (1)^2} \sqrt{(10 \times 163) - (7)^2}}$$

$$= 0.9956$$

Hence $r_{xy} = r_{uv} = 0.6$

Ex.4 Ten students got the following percentage of marks in Principles of Economics and Statistics :

Roll no. :	1	2	3	4	5	6	7	8	9	10
Marks in Economics	78	36	98	25	75	82	90	62	65	39
Marks in Statistics	84	51	91	60	68	62	86	58	53	47

calculate the coefficient of correlation.

Solⁿ Let the marks in the two subjects be denoted by x and y respectively.

x	y	$u = x - 65$	$v = y - 65$	u^2	v^2	uv
78	84	13	18	169	324	234
36	51	-29	-15	841	225	435
98	91	33	25	1089	625	825
25	60	-40	-6	1600	36	240
75	68	10	2	100	4	20
82	62	17	-4	289	16	-68
90	86	25	20	625	400	500
62	58	-3	-8	9	64	24
65	53	0	-13	0	169	0
39	47	-26	-19	676	361	494
		$\sum u = 0$	$\sum v = 0$	$\sum u^2 = 5398$	$\sum v^2 = 2224$	$\sum uv = 2734$

Here, $n=10$, $\bar{u} = \frac{1}{n} \sum u_i$, $\bar{v} = \frac{1}{n} \sum v_i = 0$

$$\begin{aligned} r_{uv} &= \frac{n \sum uv - \sum u \sum v}{\sqrt{n \sum u^2 - (\sum u)^2} \sqrt{n \sum v^2 - (\sum v)^2}} \\ &= \frac{(10 \times 2734) - (0 \times 0)}{\sqrt{(10 \times 5398) - (0)^2} \sqrt{(10 \times 2224) - (0)^2}} \\ &= 0.789 \end{aligned}$$

Hence $r_{xy} = r_{uv} = 0.789$

Ex. 5 calculate coefficient of correlation from the following results:

$n=10$, $\sum x=100$, $\sum y=150$, $\sum (x-10)^2=180$, $\sum (y-15)^2=215$

$$\sum (x-10)(y-15) = 60$$

Solⁿ Mean of first series, $\bar{x} = \frac{\sum x}{n} = \frac{100}{10} = 10$

Mean of second series, $\bar{y} = \frac{\sum y}{n} = \frac{150}{10} = 15$

$$\begin{aligned} \text{Now, } r &= \frac{\sum (x-\bar{x})(y-\bar{y})}{\sqrt{\sum (x-\bar{x})^2 \cdot \sum (y-\bar{y})^2}} \\ &= \frac{\sum (x-10)(y-15)}{\sqrt{\sum (x-10)^2 \cdot \sum (y-15)^2}} = \frac{60}{\sqrt{180 \times 215}} \\ &= 0.305 \end{aligned}$$

Rank correlation:- Sometimes we have to deal with problems in which data cannot be quantitatively measured but qualitative assessment is possible.

Let a group of n individuals be arranged in order of merit or proficiency in possession of two characteristics A and B . The rank in the two characteristics are, in general, different.

For example, if A stands for intelligence and B for beauty, it is not necessary that the most intelligent individual may be the most beautiful and vice versa. Thus an individual who is ranked at the top for the characteristic A may be ranked at the bottom for the characteristic B .

Let (x_i, y_i) , $i = 1, 2, \dots, n$ be the ranks of the n individuals in the group for the characteristics A and B respectively. Pearsonian co-efficient of correlation between the rank x_i 's and y_i 's is called the rank correlation co-efficient

between the characteristics A and B for that group of individuals.

Spearman's rank correlation coefficient, usually denoted by $\rho(Rho)$ is given by the formula

$$\rho = 1 - \frac{6 \sum d^2}{n(n^2 - 1)}$$

where d is the difference between the pair of ranks of the same individual in the two characteristics and n is the number of pairs.

Ex-1 Compute the rank correlation coefficient for the following data :

Person	A	B	C	D	E	F	G	H	I	J
Rank in maths	9	10	6	5	7	2	4	8	1	3
Rank in Physics	1	2	3	4	5	6	7	8	9	10

Solⁿ Here the ranks are given
and $n = 10$

Person	R_1	R_2	$d = R_1 - R_2$	d^2
A	9	1	8	64
B	10	2	8	64
C	6	3	3	9
D	5	4	1	1
E	7	5	2	4
F	2	6	-4	16
G	4	7	-3	9
H	8	8	0	0
I	1	9	-8	64
J	3	10	-7	49
				$\Sigma d^2 = 280$

$$r = 1 - \frac{6 \Sigma d^2}{n(n^2 - 1)}$$

$$= 1 - \frac{6 \times 280}{10(100 - 1)}$$

$$= 1 - 1.697$$

$$= -0.697$$

Ex.2 The marks secured by recruits in the Selection test (X) and in the proficiency test (Y) are given below:

Sr. No.	1	2	3	4	5	6	7	8	9
X	10	15	12	17	13	16	24	14	22
Y	30	42	45	46	33	34	40	35	39

calculate the rank correlation coefficient.

Solⁿ Here the marks are given. Therefore, first of all, write down ranks. In each series, the item with the largest size is ranked 1, next largest 2 and so on. Here $n=9$

X	10	15	12	17	13	16	24	14	22	Total
Y	30	42	45	46	33	34	40	35	39	
Ranks in X(x)	9	5	8	3	7	4	1	6	2	
Ranks in Y(y)	9	3	2	1	8	7	4	6	5	
$d = x - y$	0	2	6	2	-1	-3	-3	0	-3	0
d^2	0	4	36	4	1	9	9	0	9	72

$$\therefore r = 1 - \frac{6 \sum d^2}{n(n^2-1)}$$

$$= 1 - \frac{6 \times 72}{9 \times 80} = 1 - 0.6 = 0.4$$

Tied Ranks :- If any two or more individuals have same rank or the same value in the series of marks, then the above formula fails and requires an adjustment. In such cases, each individual is given an average rank. This common average rank is the average of the ranks which these individuals would have assumed if they were slightly different from each other. Thus, if two individual are ranked equal at the sixth place, they would have assumed the 6th and 7th ranks if they were ranked slightly different. Their common rank $= \frac{6+7}{2} = 6.5$. If three individuals are ranked equal at fourth place, they would have assumed the 4th, 5th and 6th ranks if they were rank slightly different. Their common rank $= \frac{4+5+6}{3} = 5$

Adjustment : Add $\frac{1}{12} m(m^2-1)$ to $\sum d^2$

where m stands for the number of times

an item is repeated.

This adjustment factor is to be added for each repeated item.

$$\therefore r = 1 - \frac{6 \left\{ \sum d^2 + \frac{1}{12} m_1(m_1^2 - 1) + \frac{1}{12} m_2(m_2^2 - 1) + \dots \right\}}{n(n^2 - 1)}$$

Ex. 14 Obtain the rank correlation coefficient for the following data:

X	68	64	75	50	64	80	75	40	55	64
Y	62	58	68	45	81	60	68	48	50	70

Solⁿ Here, marks are given, so write down the ranks

X	68	64	75	50	64	80	75	40	55	64	Total
Y	62	58	68	45	81	60	68	48	50	70	
Rank in X(x)	4	6	2.5	9	6	1	2.5	10	8	6	
Rank in Y(y)	5	7	3.5	10	1	6	3.5	9	8	2	
d = x - y	-1	-1	-1	-1	5	-5	-1	1	0	4	0
d ²	1	1	1	1	25	25	1	1	0	16	72

In the X-Series, the value 75 occurs twice. Had these values been slightly different, they would have been given the ranks 2 and 3. Therefore, the common rank given

to them is $\frac{2+3}{2} = 2.5$. The value 64 occurs thrice. Had these values been slightly different, they would have been given the rank 5, 6 and 7. Therefore the common rank given to them is $\frac{5+6+7}{3} = 6$.

Similarly, in the Y-series, the value 60 occurs twice. Had these value been slightly different, they would have been given the rank 3 and 4. Therefore, the common rank given to them is $\frac{3+4}{2} = 3.5$

Thus, m has the values 2, 3, 2.

$$\therefore r_i = 1 - \frac{6 \left\{ \sum d^2 + \frac{1}{12} m_1(m_1^2 - 1) + \frac{1}{12} m_2(m_2^2 - 1) + \frac{1}{12} m_3(m_3^2 - 1) \right\}}{n(n^2 - 1)}$$

$$= 1 - \frac{6 \left\{ 72 + \frac{1}{12} \cdot 2(2^2 - 1) + \frac{1}{12} \cdot 3(3^2 - 1) + \frac{1}{12} \cdot 2(2^2 - 1) \right\}}{10(10^2 - 1)}$$

$$= 1 - \frac{6 \times 75}{990}$$

$$= \frac{6}{11} = 0.545$$

Numericals for Practice

Q1 Find the coefficient of correlation from the following points of observation
 $(1,3), (2,2), (3,5), (4,4), (5,6)$.

Ans = 0.8

Q2 Calculate the coefficient of correlation between the age of husband and wife.

Age of Husband	35	34	40	43	56	20	38
Age of wife	32	30	31	32	53	20	33

Ans = 0.937

Q3 Compute the coefficient of correlation between x & y

1. Arithmetic mean of x series is 25 and that of y is 18.
2. Sum of products of deviations of x and y series from their respective means = 122.
3. Sum of squares of deviation from their respective means are 136, 138 respectively
4. Number of pairs of values = 15.

Ans = 0.89

Q4 Ten competitors in a contest are ranked by three judges in the following order

Ist Judge	1	6	5	10	3	2	4	9	7	8
IInd Judge	3	5	8	4	7	10	2	1	6	9
III Judge	6	4	9	8	1	2	3	10	5	7

Use the correlation coefficient to determine which pair of judges has the nearest approach

Ans. Ist & third Judge. $\therefore R_{13} = 0.6363, R_{12} = -0.2121, R_{23} = 0.2970$

Q5 The coefficient of rank correlation of marks obtained by 10 students in English and Economics

was found to be 0.5. It was later discovered that the difference in ranks in the two subjects obtained by one of the students was wrongly taken as 3 instead of 7. Find the correct coefficient of rank correlation.

Ans. $R = 0.2676$

Q.6 Calculate coefficient of rank correlation between X & Y.

X	78	89	97	69	59	79	68	57
Y	125	137	156	112	107	136	123	108

Ans. $R = 0.952$

Q.7 The Marks obtained by 8 students in Mathematics and Physics are given below. Compute rank correlation coefficient.

Marks in Mathematics	15	20	28	12	40	60	20	80
Marks in Physics	40	30	50	30	20	10	30	60

$R = 0$

Q.8 Calculate the rank correlation coefficient.

X	48	33	40	9	16	16	65	24	16	57
Y	13	13	24	6	15	4	20	9	6	19

$R = 0.733$

Topic-3 Regression

Regression Analysis: - Regression Analysis is a statistical technique for investigating and modelling the relationship between variables. The variable whose value is to be predicted is called dependent variable and the variable that is used for prediction is called independent variable.

Common Uses of Regression Analysis

1. **Prediction & Forecasting:** To predict future values based on past data. eg. Predicting house prices from size, location, and number of rooms.
2. **Identifying Relationships:** To understand how a change in one variable affects other. eg. How does advertising budget affect sales?
3. **Trend Analysis:** - To find long term patterns in time-series data. eg. studying how temperature changes with altitude.
4. **Risk Analysis:** - To evaluate how different factors contribute to a risk. eg. estimating the risk of heart disease from cholesterol, age, and weight.
5. **Quality Control & Optimization:** To improve processes by understanding which factors impact the output most. eg. In manufacturing, determining which machine setting minimize defects.

Types of Regression

Linear Regression: A regression model that examines the relationship between one independent variable and one dependent variable using straight line equation.

Multiple Regression: A regression model that examines the relationship between one dependent variable and two or more independent variables.

Curve of Regression. If the variables in a bivariate distribution are related, the points in the scatter diagram will be more or less concentrated around a curve, called curve of regression.

Linear Regression If the relation between two variables (dependent and independent) is linear then it is represented by a straight line, known as linear regression.

Regression Line of y on x Let the lines of regression of y on x be $y = a + bx$ ——— (1) where y is dependent variable and x is independent variable. According to the principle of least squares, we find a and b, by method of curve fitting using normal equations

$$\sum_{i=1}^n y_i = \sum_{i=1}^n a + b \sum_{i=1}^n x_i \quad \text{———— (2)}$$

$$\sum_{i=1}^n x_i y_i = a \sum_{i=1}^n x_i + b \sum_{i=1}^n x_i^2 \quad \text{———— (3)}$$

rewriting eqⁿ (2) $\frac{\sum_{i=1}^n y_i}{n} = a + b \frac{1}{n} \sum_{i=1}^n x_i$

$$\Rightarrow \bar{y} = a + b\bar{x} \quad \text{———— (4)}$$

(1)-(4) gives $y - \bar{y} = b(x - \bar{x})$ ——— (5), b is called regression coefficient of y on x, denoted as b_{yx} .

Hence, $y - \bar{y} = b_{yx}(x - \bar{x})$ is called regression line of y on x.

If r is the coefficient of correlation between x and y.

$$r = \frac{\frac{1}{n} \sum_{i=1}^n x_i y_i - \bar{x} \bar{y}}{\sigma_x \sigma_y}$$

or $r \sigma_x \sigma_y + \bar{x} \bar{y} = \frac{1}{n} \sum_{i=1}^n x_i y_i$ ——— (6)

dividing equation (3) by n , we get

$$\frac{1}{n} \sum_{i=1}^n x_i y_i = \frac{a \sum_{i=1}^n x_i}{n} + b \frac{\sum_{i=1}^n x_i^2}{n}$$

$$\text{or } \frac{1}{n} \sum_{i=1}^n x_i y_i = a\bar{x} + b \frac{1}{n} \sum_{i=1}^n x_i^2 \quad \text{--- (7)}$$

using eqⁿ (6)

$$r\sigma_x\sigma_y + \bar{x}\bar{y} = a\bar{x} + \frac{1}{n} \sum_{i=1}^n x_i^2 \quad \text{--- (8)}$$

Multiply by $\bar{x}$ in eqⁿ (4), we get

$$\bar{x}\bar{y} = a\bar{x} + b\bar{x}^2 \quad \text{--- (9)}$$

subtracting eqⁿ (9) from (8)

$$r\sigma_x\sigma_y = b \cdot \frac{1}{n} \sum_{i=1}^n x_i^2 - b\bar{x}^2$$

$$\text{or } r\sigma_x\sigma_y = b\sigma_x^2$$

$$b_{yx} = \frac{r\sigma_x\sigma_y}{\sigma_x^2} = r \frac{\sigma_y}{\sigma_x}$$

Hence $y - \bar{y} = \frac{r\sigma_y}{\sigma_x} (x - \bar{x})$ is equation of line of regression of y on x .

Regression Line of x on y : Let $x = a + by$ is eqⁿ of regression line of x on y , where x is dependent variable and y is independent variable.

$x - \bar{x} = b_{xy}(y - \bar{y})$, where $b_{xy} = r \frac{\sigma_x}{\sigma_y}$ is called regression line of x on y .

Remark:- Two lines of regression pass through the point $(\bar{x}, \bar{y})$.

Remark:- Another form of regression coefficients

$$b_{yx} = \frac{n \sum xy - \sum x \sum y}{n \sum x^2 - (\sum x)^2}$$

$$b_{xy} = \frac{n \sum xy - \sum x \sum y}{n \sum y^2 - (\sum y)^2}$$

Properties of Regression Lines.

4

1. The correlation coefficient is geometric mean between regression coefficients.

$$\text{since } b_{yx} = \frac{r\sigma_y}{\sigma_x}, \quad b_{xy} = \frac{r\sigma_x}{\sigma_y}$$
$$b_{yx} \times b_{xy} = r^2 \Rightarrow r = \sqrt{b_{yx} \times b_{xy}}$$

2. If one of the regression coefficient is greater than unity, the other must be less than unity.
3. Both regression coefficient & correlation coefficient must have same sign.
4. Arithmetic mean of regression coefficients is greater than the correlation coefficient.
5. If x & y are independent, the regression coefficients are zero.
6. The product of regression coefficients is always less than or equal to 1.

Angle between two regression lines.

Q.1 If θ is an angle between two regression lines show that, $\tan \theta = \frac{1-r^2}{r} \frac{\sigma_x \sigma_y}{\sigma_x^2 + \sigma_y^2}$.

Interpret the cases when $r=0$, $r=\pm 1$.

Proof. The eqⁿs of two lines of regression are

$$y - \bar{y} = \frac{r\sigma_y}{\sigma_x} (x - \bar{x}) \quad \text{--- (1)}$$

$$(x - \bar{x}) = \frac{r\sigma_x}{\sigma_y} (y - \bar{y}) \quad \text{--- (2)}$$

The slopes of these lines of regression are

$$m_1 = \frac{r\sigma_y}{\sigma_x}, \quad m_2 = \frac{1}{r} \frac{\sigma_y}{\sigma_x}$$

The angle θ between two lines is $\tan \theta = \frac{m_2 - m_1}{1 + m_1 m_2}$

$$\Rightarrow \tan \theta = \frac{\frac{\sigma_y}{r\sigma_x} - \frac{r\sigma_y}{\sigma_x}}{\frac{1 + \frac{r\sigma_y}{\sigma_x} \times \frac{\sigma_y}{r\sigma_x}}{\frac{1 + \frac{\sigma_y^2}{\sigma_x^2}}{\frac{\sigma_x^2}{\sigma_y^2}}}} = \frac{\frac{\sigma_y}{\sigma_x} \left(\frac{1-r^2}{r} \right)}{\frac{1 + \frac{\sigma_y^2}{\sigma_x^2}}{\frac{\sigma_x^2}{\sigma_y^2}}} = \left(\frac{1-r^2}{r} \right) \frac{\sigma_x \sigma_y}{\sigma_x^2 \sigma_y^2}$$

Hence proved.

Case-I When $r=0$, we get $\tan \theta = \infty$
 $\Rightarrow \theta = \pi/2$

i.e. two lines of regression are perpendicular to each other.

Case-II If $r = \pm 1$, we get $\tan \theta = 0 \Rightarrow \theta = 0$
 i.e. two lines of regression coincide and there is perfect correlation between the variables x and y .

Numericals

Q1 Two regression lines are $3x+2y-26=0$ and $6x+y-31=0$
 Calculate

- 1) mean $\bar{x}$ and $\bar{y}$
- 2) Correlation coefficient
- 3) Variance of y , if variance of x is 25.

Sol. 1) Since $(\bar{x}, \bar{y})$ is a common point of the two lines of regression, we have

$$3\bar{x} + 2\bar{y} = 26$$

$$6\bar{x} + \bar{y} = 31$$

on solving, we get $\bar{x}=4, \bar{y}=7$

2) Since $r = \sqrt{b_{yx} \cdot b_{xy}}$

Let regression line of y on x is $y = 13 - \frac{3}{2}x$
 $\Rightarrow b_{yx} = -\frac{3}{2}$

Regression line of x on y is $x = -\frac{y}{6} + \frac{31}{6}$

$$b_{xy} = -\frac{1}{6}$$

$$r = \sqrt{-\frac{3}{2} \times -\frac{1}{6}} = \sqrt{\frac{1}{4}} = \pm \frac{1}{2}$$

3) variance of $x = 25 \Rightarrow \sigma_x^2 = 25 \Rightarrow \sigma_x = 5$

$\therefore b_{yx} = r \frac{\sigma_y}{\sigma_x} = \frac{-3}{2} = -\frac{1}{2} \frac{\sigma_y}{5} \Rightarrow \sigma_y = 15$

$\therefore$ variance of $y = \boxed{\sigma_y^2 = 225}$

Q. In a partially destroyed laboratory record of an analysis of correlation data, the following results are eligible:

variance of $x = 9$, Regression equations: $8x - 10y + 66 = 0$,
 $40x - 18y = 214$.

What were a) the mean values of x and y .
 b) the standard deviation of y
 c) coefficient of correlation between x and y ?

Sol. 9) Since both regression lines pass through the point $(\bar{x}, \bar{y})$, therefore, we have

$$8\bar{x} - 10\bar{y} = -66$$

$$40\bar{x} - 18\bar{y} = 214$$

on solving, we get

$$\bar{x} = 13, \bar{y} = 17$$

5) consider regression line of y on x

$$10y = 8x + 66$$

$$y = \frac{4}{5}x + \frac{33}{5} \quad \text{--- (1)}$$

regression line of x on y .

$$x = \frac{9}{20}y + \frac{107}{20} \quad \text{--- (2)}$$

$$b_{yx} = \frac{4}{5}, \quad b_{xy} = \frac{9}{20}$$

$$r = \sqrt{b_{yx} b_{xy}} = \sqrt{\frac{4}{5} \times \frac{9}{20}} = \pm \frac{3}{5}$$

$\therefore b_{yx}, b_{xy} > 0 \Rightarrow \boxed{r = \frac{3}{5}}$

b) $\therefore \sigma_x^2 = 9 \Rightarrow \sigma_x = 3$

$b_{yx} = r \frac{\sigma_y}{\sigma_x} \Rightarrow \frac{4}{5} = \frac{3}{5} \frac{\sigma_y}{3} \Rightarrow \sigma_y = 4$
 standard deviation of $y = 4$

Q. Obtain the equations of the lines of regression for the following data:

x: 1 2 3 4 5 6 7 8 9
y: 9 8 10 12 11 13 14 16 15

Also obtain an estimate of y for x=6.2.

x	y	x^2	y^2	xy
1	9	1	81	9
2	8	4	64	16
3	10	9	100	30
4	12	16	144	48
5	11	25	121	55
6	13	36	169	78
7	14	49	196	98
8	16	64	256	128
9	15	81	225	135
45	108	285	1356	597

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n} = \frac{45}{9} = 5$$

$$\bar{y} = \frac{\sum_{i=1}^n y_i}{n} = \frac{108}{9} = 12$$

$$\text{Regression coefficient } b_{yx} = \frac{n \sum xy - \sum x \sum y}{n \sum x^2 - (\sum x)^2}$$

$$b_{xy} = \frac{n \sum xy - \sum x \sum y}{n \sum y^2 - (\sum y)^2}$$

$$\text{we get, } b_{yx} = \frac{9(597) - 45 \times 108}{9 \times 285 - (45)^2} = 0.95$$

$$b_{xy} = \frac{9 \times 597 - 45 \times 108}{9 \times 1356 - (108)^2} = 0.95$$

Eqⁿ of line of regression of y on x is

$$y - \bar{y} = b_{yx} (x - \bar{x}) \Rightarrow (y - 12) = 0.95 (x - 5)$$

$$\Rightarrow y = 0.95x + 7.25$$

Estimate of y for $x = 6.5$

$$y = 0.95 \times 6.5 + 7.25 = 13.14$$

The eqⁿ of line of regression of x on y is

$$(x - \bar{x}) = b_{xy} (y - \bar{y}) \Rightarrow (x - 5) = 0.95 (y - 12)$$

$$\Rightarrow x = 0.95y - 6.4$$

Q. The following data regarding the heights (y) and weights (x) of 100 college students is given:

$$\sum x = 15000; \quad \sum x^2 = 2272500; \quad \sum y = 6800$$

$$\sum y^2 = 463025; \quad \sum xy = 1022250$$

Find the correlation coefficient between height and weight and state the equation of regression of height on weight.

Sol. Correlation coefficient, $r = \frac{n \sum xy - \sum x \sum y}{\sqrt{n \sum x^2 - (\sum x)^2} \sqrt{n \sum y^2 - (\sum y)^2}}$

Here $n = 100$,

$$\Rightarrow r = \frac{100 \times 1022250 - 15000 \times 6800}{\sqrt{100 \times 2272500 - (15000)^2} \sqrt{100 \times 463025 - (6800)^2}}$$

$$\Rightarrow \frac{225000}{\sqrt{2250000} \sqrt{62500}} = \frac{225000}{1500 \times 250} = 0.6$$

$$r = 0.6$$

Regression eqⁿ of height (y) on weight (x) is

$$y - \bar{y} = b_{yx} (x - \bar{x})$$

$$b_{yx} = \frac{n \sum xy - \sum x \sum y}{n \sum x^2 - (\sum x)^2} = \frac{100 \times 1022250 - 15000 \times 6800}{100 \times 2272500 - (15000)^2} = \frac{1}{10}$$

$$\therefore (y-68) = \frac{1}{10}(x-150) \Rightarrow 10y-680 = x-150$$

Hence, $\boxed{y = \frac{x}{10} + 53}$ Ans.

Q. The following results were obtained from records of age (x) and systolic blood pressure (y) of a group of 10 men;

Mean $\bar{x} = 53$, $\bar{y} = 142$ and $\sum(x-\bar{x})(y-\bar{y}) = 1220$.
Variance $\sigma_x^2 = 130$

Find the appropriate regression equation and use it to estimate the blood pressure of a man whose age is 45.

Sol. To estimate blood pressure (y) of a 45 years old man, we will find regression line of y on x .

$$y - \bar{y} = b_{yx}(x - \bar{x})$$

$$b_{yx} = \frac{\lambda \sigma_y}{\sigma_x} = \frac{\sum(x-\bar{x})(y-\bar{y})}{n \sigma_x \sigma_y} \left(\frac{\sigma_y}{\sigma_x} \right) = \frac{\sum(x-\bar{x})(y-\bar{y})}{n \sigma_x^2}$$

$$= \frac{1220}{10(130)} = 0.93846$$

$$\therefore y - \bar{y} = 0.93846(x - 53)$$

$$\text{or } \boxed{y = 0.93846x + 92.26162}$$

When $x = 45$, $y = 134.49$ Ans.

Q. The following data gives the correlation coefficient, means and s.d. of rainfall and yield of paddy in a certain tract.

	Yield per acre (in lbs.)	Annual rainfall (in inches)
Mean	972.2	16.5
S.D.	40.3	2.2

coefficient of correlation 0.55. Estimate the most likely yield of paddy when the annual rainfall is 20", other factors remain the same.

Sol. Let y be the variable for yield and x for annual rainfall.

The most likely yield of paddy when rainfall is 20".

consider regression line of y on x , $y - \bar{y} = b_{yx}(x - \bar{x})$

Here $\bar{x} = 16.5$, $\sigma_x = 2.2$, $\bar{y} = 972.2$, $\sigma_y = 40.3$, $r = 0.55$.

$$b_{yx} = \frac{r\sigma_y}{\sigma_x} = \frac{(0.55)(40.3)}{2.2} = 10.075$$

$\therefore$ equation of line of regression of y on x is

$$y - 972.2 = 10.075(x - 16.5)$$

$$\Rightarrow y = 10.075x + 805.96$$

The estimate of most likely yield of paddy when annual rainfall is 20", given by

$$y = 10.075 \times 20 + 805.96 = \boxed{1007.4625} \text{ Ans.}$$

Numericals for Practice.

Q.1 Find the correlation coefficient and the equations of regression lines for the following values of x and y :

x :	1	2	3	4	5
y :	2	5	3	8	7

Ans. $r = 0.81$, $x = 0.5y + 0.5$, $y = 1.3x + 1.1$.

If the two regression coefficients are 0.8 and 0.2, find the value of coefficient of correlation.

Ans. 0.4.

Q.3 Find the most likely price in Bombay corresponding to the price of Rs 70 at Calcutta from the following:

	Calcutta	Bombay
Average price	65	67
S.D.	2.7	3.5

Correlation coefficient between the prices of commodities in the two cities is 0.8.

Ans.

Q.4 An analyst for a company was studying travelling allowance (y) in ₹ and duration (x) of these trips

for 102 sales trip. He has found relation between x and y linear and data as follows.

$$\sum x = 510, \sum y = 7140, \sum x^2 = 4150, \sum xy = 54900, \sum y^2 = 74200$$

calculate

- Two regression lines.
- A given trip has to take 7 days. How much money should be allowed so that they will not run short of money?

Ans.

Q. Assuming that we conduct an experiment with 8 fields planted with corn, four fields having no nitrogen fertilizer and four fields having 80 kg of nitrogen fertilizer. The resulting corn yields are shown in table in bushels per acre:

Field:	1	2	3	4	5	6	7	8
Nitrogen x : (kg)	0	0	0	0	80	80	80	80
Corn yield y : (acre)	120	360	60	180	1280	1120	1120	760

- Compute a linear regression equation of y on x .
- Predict corn yield for a field treated with 60 kg of fertilizer.

Ans

Q. If the coefficient of correlation between two variables x and y is 0.5 and the acute angle between their regression lines is $\tan^{-1}(\frac{3}{5})$, show that $\sigma_x = \frac{1}{2} \sigma_y$.

Non Linear Regression:-

While linear regression is effective for relationships that follow a straight line pattern, many real world problems exhibit more complex behaviors. In such cases, linear models fail to capture the true nature of data. Example: growth rates, decay processes, epidemics etc.

Non Linear Regression (Polynomial fit)

Nonlinear Regression is a statistical method for modelling data where relationship between dependent and independent variables is a curve or nonlinear function.

Polynomial regression is a type of nonlinear regression that fits a polynomial function to the data.

Let $y = a + bx + cx^2$ be a second degree parabolic curve of regression of y on x to be fitted for the data (x_i, y_i) , $i = 1, 2, \dots, n$.

Then, By principle of least squares, the best values of a, b and c ,

$$\sum y = na + b \sum x + c \sum x^2 \quad \text{--- (1)}$$

$$\sum xy = a \sum x + b \sum x^2 + c \sum x^3 \quad \text{--- (2)}$$

$$\sum x^2 y = a \sum x^2 + b \sum x^3 + c \sum x^4 \quad \text{--- (3)}$$

Equation (1), (2) and (3) are normal equations for fitting a second degree parabolic curve of regression of y on x .

Q. The velocity V of a liquid is known to vary with temperature T , according to a quadratic law $V = a + bT + cT^2$. Find the best values of a, b and c for following data:

T	1	2	3	4	5	6	7
V	2.31	2.01	1.80	1.66	1.55	1.47	1.41

Normal equations to fit a quadratic eqn

$$V = a + bT + cT^2 \text{ are}$$

$$\sum V = na + b \sum T + c \sum T^2 \quad \text{--- (1)}$$

$$\sum VT = a \sum T + b \sum T^2 + c \sum T^3 \quad \text{--- (2)}$$

$$\sum VT^2 = a \sum T^2 + b \sum T^3 + c \sum T^4 \quad \text{--- (3)}$$

V	T	T ²	T ³	T ⁴	VT	VT ²
2.31	1	1	1	1	2.31	2.31
2.01	2	4	8	16	4.02	8.04
1.80	3	9	27	81	5.40	16.20
1.66	4	16	64	256	6.64	26.56
1.55	5	25	125	625	7.75	38.75
1.47	6	36	216	1296	8.82	52.92
1.41	7	49	243	1701	9.87	69.09
12.21	28	140	784	4676	44.81	213.87

We get

$$12.21 = 7a + 28b + 140c$$

$$44.81 = 28a + 140b + 784c$$

$$213.87 = 140a + 784b + 4676c$$

on solving, we get $a = 2.59$, $B = -0.32$, $C = 0.02$

$$\therefore V = 2.59 - 0.32T + 0.02T^2 \text{ is non-linear}$$

regression curve.

Q. Fit a second degree polynomial from the following data:

x	0	1	2	3	4	5	6	7	8
y	12.0	10.5	10	8	7	8	7.5	8.5	9

Also estimate y at $x = 6.5$

$$\text{Ans. } y = 12.1848 - 1.8465x + 0.1829x^2, \quad 7.91$$

Multiple Regression Analysis:

In multiple regression, the objective is to build a model that relates a dependent variable y to more than one independent variable. Let k represent the number of independent variables ($k \geq 2$) denoted as $x_1, x_2, \dots, x_k$.

For example, to predict the selling price of a house, with $x_1 = \text{size (m}^2\text{)}$, $x_2 = \text{age (years)}$, $x_3 = \text{number of rooms}$.

Consider a linear function

$$y = a + bx_1 + cx_2 + dx_3$$

In general, consider a linear function as

$y = a + bx + cz$, where y is dependent variable and x, z are two independent variables.

Then, by method of least squares, Normal eq^s are.

$$\sum y = na + b\sum x + c\sum z$$

$$\sum xy = a\sum x + b\sum x^2 + c\sum xz$$

$$\sum zy = a\sum z + b\sum xz + c\sum z^2$$

on solving these three eq^s, we get the values of a, b, c . Consequently we get the linear function.

$y = a + bx + cz$ called regression plane.

Q. Obtain a regression plane by using multiple linear regression to fit the data:

x	1	2	3	4
y	0	1	2	3
z	12	18	24	30

Sol. let $y = a + bx + cz$ be the required regression plane, where a, b, c are constants.

Normal eqⁿs are

$$\begin{aligned}\sum y &= na + b \sum x + c \sum z \\ \sum xy &= a \sum x + b \sum x^2 + c \sum xz \\ \sum yz &= a \sum z + b \sum xz + c \sum z^2\end{aligned}$$

Here $n=4$

x	z	y	x^2	z^2	xy	xz	yz
1	0	12	1	0	12	0	0
2	1	18	4	1	36	2	18
3	2	24	9	4	72	6	48
4	3	30	16	9	120	12	90
$\sum x=10 \quad \sum z=6 \quad \sum y=84 \quad \sum x^2=30 \quad \sum z^2=14 \quad \sum xy=240 \quad \sum xz=20 \quad \sum yz=156$							

we get

$$\begin{aligned}84 &= 4a + 10b + 6c \\ 240 &= 10a + 30b + 20c \\ 156 &= 6a + 20b + 14c\end{aligned}$$

on solving, we get $a=10, b=2, c=4$

Hence required regression plane is $y = 10 + 2x + 4z$.

Q. An investigation of a die-casting process resulted in the accompanying data on x_1 = furnace temperature, x_2 = die close time, and y = temperature difference on the die surface. Obtain a regression plane for temperature difference.

x_1	1250	1300	1350	1250	1300	1250	1300	1350	1350
x_2	6	7	6	7	6	8	8	7	8
y	80	95	101	85	92	87	96	106	108

Ans. $y = -200 + 0.210x_1 + 3.00x_2$

CURVE FITTING

Introduction:

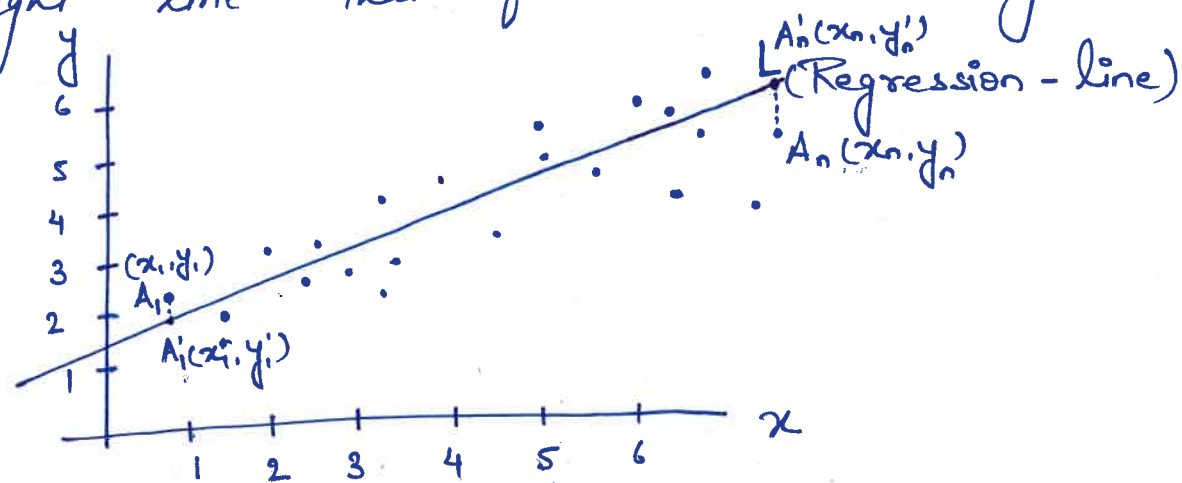
In statistics, we come across a number of situations where we need to find an association between two or more variables. For example, we sometimes need to associate demand and supply of goods, income & expenditure of a company or weight and height of a person. These associations in general may be represented in various mathematical forms such as - polynomial, exponential, logarithmic equations. In order to determine the nature of such relations, the first essential step is to collect the data showing the corresponding values of the variables under consideration. The mathematical expression can then be obtained by the method of Curve fitting.

Curve-fitting is the process of finding the 'best-fit' curve for a given set of data. It is the representation of the relationship between 2 variables by means of an algebraic equation. Among various methods of 'Curve-fitting' the most preferred one is 'Method of Least Squares'.

Method of Least Squares:

The method of least-squares assumes that the best-fit curve of a given type is the curve that has the minimum sum of squares of deviations from a given set of data.

For better elucidation, let us consider the case of a straight line. Suppose we have a set of n data points $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$ which are paired values of 2 variables say x & y . Suppose we have to find a straight line that fits best to the given data.



According to the method of least square, the line which is at minimum vertical distance from each point of data set is the line of best fit. To estimate this line, let the vertical distance of this line from the corresponding point be $y_i - \hat{y}_i$ $1 \leq i \leq n$. (If X_1 has coordinates (x_1, y_1) and the corresponding point on line which is at minimum vertical distance from (x_1, y_1) is X_1'

with co-ordinates (x_i, y_i') then the vertical distance is $(y_i - y_i')$. Now, consider the quantity

$$U = \sum (y_i - y_i')^2$$

In order to get line of best fit U has to be minimum & correspondingly each of the quantities $(y_i - y_i')$ has to be minimum.

Thus, for the straight line $y = a + bx$ a & b should be such that U is minimum

$$\text{i.e. } \frac{\partial U}{\partial a} = \frac{\partial U}{\partial b} = 0.$$

$$\text{Now, } U = \sum (y_i - y_i')^2 = \sum (y_i - (a + bx_i))^2$$

$$\frac{\partial U}{\partial a} = 0 \Rightarrow -2 \sum (y_i - (a + bx_i)) = 0$$

$$\Rightarrow \sum y_i = na + b \sum x_i \quad \text{--- (1)}$$

$$\frac{\partial U}{\partial b} = 0 \Rightarrow -2 \sum x_i (y_i - (a + bx_i)) = 0$$

$$\Rightarrow \sum x_i y_i = a \sum x_i + b \sum x_i^2 \quad \text{--- (2)}$$

Equation (1) & (2) are called Normal Equations. Solving the normal equations, we get the values of a & b which when substituted in the general equation of the line gives the equation of the straight line that fits best to the given data.

Using the similar technique, one can fit any polynomial of desired degree to the given data.

Fitting of a Straight Line:

Let we have n sets of observations say (x_1, y_1) , (x_2, y_2) , ..., (x_n, y_n) to fit in a straight line $y = a + bx$. Then, by the method of least squares, we form the normal equations:

$$\Sigma y = na + b \Sigma x \text{ --- (1)}$$

$$\Sigma xy = a \Sigma x + b \Sigma x^2 \text{ --- (2)}$$

Solving them, we get the values of a & b which when put in the general form, gives us the eq. of the straight line that fits best to the given data.

Remember that the no. of normal equations is always equal to the no. of constants to be evaluated.

Example By the method of least squares, fit a straight line that fits best to the following data:

x	1	2	3	4	5
y	14	27	40	55	68

Solution: Let the straight line of best fit be $y = a + bx$ --- (1)

Normal Equations are: $\Sigma y = na + b \Sigma x$ --- (2)

$$\Sigma xy = a \Sigma x + b \Sigma x^2 \text{ --- (3)}$$

$$n=5$$

x	y	xy	x^2
1	14	14	1
2	27	54	4
3	40	120	9
4	55	220	16
5	68	340	25
$\Sigma x = 15$	$\Sigma y = 204$	$\Sigma xy = 748$	$\Sigma x^2 = 55$

Substituting in ② & ③ we get

$$204 = 5a + 15b$$

$$748 = 15a + 55b$$

Solving, we get $a=0$, $b=13.6$

Hence the required equation of straight line is

$$y = 13.6x$$

Example

fit a straight line to the following data:

x	0	1	2	3	4
y	1	1.8	3.3	4.5	6.3

Solution:

Let the straight line of best fit be

$$y = a + bx$$

— ①

Normal

Equations are:

$$\Sigma y = na + b \Sigma x \text{ — ②}$$

$$\Sigma xy = a \Sigma x + b \Sigma x^2 \text{ — ③}$$

x	y	xy	x^2
0	1	0	0
1	1.8	1.8	1
2	3.3	6.6	4
3	4.5	13.5	9
4	6.3	25.2	16
$\Sigma x = 10$	$\Sigma y = 16.9$	$\Sigma xy = 47.1$	$\Sigma x^2 = 30$

From (2) & (3) $16.9 = 5a + 10b$ — (4)

$47.1 = 10a + 30b$ — (5)

Solving (4) & (5) we get $a = 0.72$, $b = 1.33$

Hence the equation of required straight line is

$$y = 0.72 + 1.33x$$

Fitting of a quadratic Equation or fitting of a parabola

The general equation for a parabola or the general quadratic equation / equation of second degree is:

$$y = a + bx + cx^2$$

By the method of least square, the corresponding normal equations will be:

$$\Sigma y = na + b\Sigma x + c\Sigma x^2 \text{ --- (1)}$$

$$\Sigma xy = a\Sigma x + b\Sigma x^2 + c\Sigma x^3 \text{ --- (2)}$$

$$\Sigma x^2y = a\Sigma x^2 + b\Sigma x^3 + c\Sigma x^4 \text{ --- (3)}$$

Solving these equations with the help of provided data, one can get the values of constants which on putting in the general equation, gives the desired equation.

Example: Fit a parabola to the following data:

x	-1	0	1	3
y	6	3	4	18

Solution: The general equation of parabola is: $y = a + bx + cx^2$ --- (1)

The corresponding normal equations are:

$$\Sigma y = na + b\Sigma x + c\Sigma x^2 \text{ --- (2)}$$

$$\Sigma xy = a\Sigma x + b\Sigma x^2 + c\Sigma x^3 \text{ --- (3)}$$

$$\Sigma x^2y = a\Sigma x^2 + b\Sigma x^3 + c\Sigma x^4 \text{ --- (4)}$$

x	y	x ²	x ³	x ⁴	xy	x ² y
-1	6	1	-1	1	-6	6
0	3	0	0	0	0	0
1	4	1	1	1	4	4
3	18	9	27	81	54	162
3	31	11	27	83	52	172

$$4a + 3b + 11c = 31 \text{ --- } ⑤$$

$$3a + 11b + 27c = 52 \text{ --- } ⑥$$

$$11a + 27b + 83c = 172 \text{ --- } ⑦$$

Solving, we get $a = 3$, $b = -1$, $c = 2$

Substituting in equation ①,

$$y = 3 - x + 2x^2$$

Example: By the method of least squares, fit the curve $y = ax + bx^2$ that best fits the following data:

	1	2	3	4	5
x					
y	1.8	5.1	8.9	14.1	19.8

Solution: The given equation is:

$$y = ax + bx^2$$

Normal Equations are:

$$\sum xy = a\sum x + b\sum x^2$$

$$\sum x^2 y = a\sum x^2 + b\sum x^3$$

Solving these, we get A & B .

Then, $a = \text{Antilog } A$ & $b = \frac{B}{\log_{10} e}$

Fitting of the curve $y = ab^x$

The given equation is $y = ab^x$

Taking logarithm on both sides,

$$\log_{10} y = \log_{10} a + x \log_{10} b$$

Let $Y = \log_{10} y$, $A = \log_{10} a$, $X = x$, $B = \log_{10} b$

$$\Rightarrow \boxed{Y = A + BX}$$

We find A & B by making normal equations and solving them.

Then, $a = \text{Antilog}(A)$ & $b = \text{Antilog}(B)$

Fitting of the curve $y = ax^b$

The given equation is $y = ax^b$

Taking logarithm on both sides

$$\log_{10} y = \log_{10} a + b \log_{10} x$$

Let $\log_{10} y = Y$, $\log_{10} a = A$, $b = B$, $\log_{10} x = X$

$$\Rightarrow \boxed{Y = A + BX}$$

x	y	x^2	x^3	x^4	xy	x^2y
1	1.8	1	1	1	1.8	1.8
2	5.1	4	8	16	10.2	20.4
3	8.9	9	27	81	26.7	80.1
4	14.1	16	64	256	56.4	225.6
5	19.8	25	125	625	99	495
		55	225	979	194.1	822.9

$$55a + 225b = 194.1$$

$$225a + 979b = 822.9$$

Solving, $a = 1.52$, $b = 0.49$

$$\Rightarrow y = 1.52x + 0.49x^2$$

Fitting of an exponential Curve $y = ae^{bx}$

The given equation is $y = ae^{bx}$

Take logarithm on both sides

$$\log_{10} y = \log_{10} a + bx \log_{10} e$$

Let $\log_{10} y = Y$, $\log_{10} a = A$, $b \log_{10} e = B$, $x = X$

$$\Rightarrow \boxed{Y = A + BX}$$

Corresponding normal equations are:

$$\sum Y = nA + B \sum X \quad \text{--- ①}$$

$$\sum XY = A \sum X + B \sum X^2 \quad \text{--- ②}$$

We find A & B by making normal equations and then solving them.

Then, $a = \text{Antilog}(A)$ & $b = B$

Example: Find the curve of best fit of the type $y = ae^{bx}$ to the following data by the method of least squares:

x	1	5	7	9	12
y	10	15	12	15	21

Solution: The given equation is

$$y = ae^{bx}$$

$$\Rightarrow y = A + Bx \text{ where } y = \log_{10} y, A = \log_{10} a,$$

$$x = x \text{ and } B = b \log_e e$$

The normal equations are

$$\sum y = 5A + B \sum x \text{ — ①}$$

$$\sum xy = A \sum x + B \sum x^2 \text{ — ②}$$

$X = x$	y	$y = \log_{10} y$	x^2	xy
1	10	1.0000	1	1
5	15	1.1761	25	5.8805
7	12	1.0792	49	7.5544
9	15	1.1761	81	10.5849
12	21	1.3222	144	15.8664
$\sum x = 34$		$\sum y = 5.7536$	$\sum x^2 = 300$	$\sum xy = 40.8862$

$X = x$	y	$y = \log_{10} y$	X^2	XY
2	8.3	0.9191	4	1.8382
3	15.4	1.1872	9	3.5616
4	33.1	1.5198	16	6.0792
5	65.2	1.8142	25	9.0710
6	127.4	2.1052	36	12.6312
$\Sigma X = 20$		$\Sigma Y = 7.5455$	$\Sigma X^2 = 90$	$\Sigma XY = 33.1812$

Substituting in normal equations,

$$5A + 20B = 7.5455$$

$$20A + 90B = 33.1812$$

Solving, $A = 0.31$, $B = 0.3$

$\Rightarrow a = \text{Antilog } A = 2.04$ & $b = \text{Antilog } B = 1.995$

$\Rightarrow$ The required curve is $y = 2.04 (1.995)^x$

Example: By using the method of least square, find a relation of the form $y = ax^b$ that fits the data:

x	2	3	4	5
y	27.8	62.1	110	161

Solution: The curve to be fitted is

$$y = ax^b$$

$$\text{or, } Y = A + BX$$

where $Y = \log_{10} y$, $A = \log_{10} a$, $X = \log_{10} x$, $B = b$

Substituting the values in the normal equations,

$$5A + 34B = 5.7536$$

$$34A + 300.8B = 40.8862$$

Solving, $A = 0.9766$, $B = 0.02561$

$$\Rightarrow a = \text{Antilog}_{10} A = 10^A = 9.4754$$

$$b = \frac{B}{\log_{10} e} = 0.059$$

Hence the required curve is

$$y = 9.4754 e^{0.059x}$$

Example: Obtain a relation of the form

$y = ab^x$ for the following data:

x	2	3	4	5	6
y	8.3	15.4	33.1	65.2	127.4

Solution: The curve to be fitted is

$$y = ab^x$$

OR, $Y = A + BX$

where, $Y = \log_{10} y$, $A = \log_{10} a$, $X = x$, $B = \log_{10} b$

Normal Equations are, $\sum Y = 5A + B\sum X$
 $\sum XY = A\sum X + B\sum X^2$

Normal Equations are

$$\sum y = 4A + B \sum x$$

$$\sum xy = A \sum x + B \sum x^2$$

x	y	$X = \log_{10} x$	$Y = \log_{10} y$	X^2	XY
2	27.8	0.3010	1.440	0.0906	0.4346
3	62.1	0.4771	1.7931	0.2276	0.8555
4	11.0	0.6021	2.0414	0.3625	1.2291
5	161	0.6990	2.2068	0.4886	1.5426
		$\sum X = 2.0792$	$\sum Y = 7.4853$	$\sum X^2 = 1.1693$	$\sum XY = 4.0618$

$$4A + 2.0792B = 7.4853$$

$$2.0792A + 1.1693B = 4.0618$$

Solving, we get $A = 0.8678$ $B = 1.9311$

$\Rightarrow a = \text{Antilog } A = 7.376$, $b = B = 1.9311$

$\Rightarrow$ The required curve is

$$y = 7.376x^{1.9311}$$

Example: Electromagnetic Technologies offer effective non-destructive sensing techniques for determining characteristics of pavement. The propagation of electromagnetic waves through the material depends on its dielectric properties. The following data has to be used to relate $y = \text{dielectric constant}$ to $x = \text{air void}$. Fit a line to the data using the method of least square:

x	1	2	3	4	5	6	7	8
y	1	1.2	1.8	2.5	3.6	4.7	6.6	9.1

Solution: General Equation: $y = a + bx$

Normal Equations:
$$\begin{aligned} 3 \sum y &= na + b \sum x \\ \sum xy &= a \sum x + b \sum x^2 \end{aligned}$$

Upon finding summations,
$$\begin{aligned} 30.5 &= 8a + 36b \\ 184 &= 36a + 204b \end{aligned}$$

Solving, $a = 1.10714$, $b = -1.16963$

⇒ Required relation between x and y

$$y = 1.10714x - 1.16963$$

Example: Visual and musculoskeletal problems associated with the use of visual display terminals (VDTs) have become rather common in recent years. Some researchers have focused on vertical gaze direction as a source of eye strain and irritation. Using the following data for x = width of palpebral fissure and y = OSA (Ocular Surface Area), establish an exponential relation $y = ae^{bx}$ between the given variables:

x	2	4	6	8	10
y	4.077	11.084	30.128	81.897	222.62

Solution: Given equation is $y = ae^{bx}$
 $\Rightarrow Y = A + BX$ where $Y = \log y$, $A = \log a$,
 $B = b \log e$, $X = \log x$

Normal Equations: $\sum Y = nA + B \sum X$
 $\sum XY = A \sum X + B \sum X^2$

Upon finding summations,

$$7.394824 = 5A + 30B$$

$$53.05498 = 30A + 220B$$

$$\text{Solving, } A = 0.1760594, B = 0.2171509$$

$$a = \text{Antilog}(A) = 1.49989, b = \frac{B}{\log e} = 0.50001$$

$$\Rightarrow y = 1.49989 e^{0.50001x}$$

Example: Arsenic is found in many ground waters and some surface waters. Recent health effects research has prompted the Environment Protection Agency to reduce allowable arsenic levels in drinking water so that many water systems are no longer compliant with standards. This has spurred interest in the development of methods to remove arsenic. The accompanying data on $x = \text{pH}$ and $y = \text{arsenic removed percentage}$ by a particular process is provided to fit a quadratic relation $y = a + bx + cx^2$ between the variables.

x	1	2	3	4	5	6	7
y	2.31	2.01	1.80	1.66	1.55	1.47	1.41

Solution: $y = 2.59477 - 0.32632x + 0.02274x^2$

Q: Find the least square line $y = a + bx$ for the data:

x	-2	-1	0	1	2
y	1	2	3	3	4

Ans: $y = 2.6 + (0.7)x$

Q: Find the least square fit of the form $y = a_0 + a_1x^2$ to the following data:

x	-1	0	1	2
y	2	5	3	0

Ans: $y = 4.1667 - 1.1111x^2$

Q: Fit $y = ax^2 + bx$ to the following data:

x	1	2	3	4	5	6	7	8
y	1	1.2	1.8	2.5	3.6	4.7	6.6	9.1

Ans: $y = 0.107798x^2 + 0.217125x$

Q: Fit a second degree parabola in the following data:

x	0	1	2	3	4
y	1	4	10	17	30

Ans: $y = 1.2 + 1.1x + 1.5x^2$

Q: For the data given below, find the equation to the best fitting exponential curve of the form $y = ae^{bx}$

x	1	2	3	4	5	6
y	1.6	4.5	13.8	40.2	125	300

Ans: $y = 0.5580e^{1.0631x}$

Q: Fit a least square geometric curve $y = ax^b$ to the following data:

x	1	2	3	4	5
y	0.5	2	4.5	8	12.5

Ans: $y = 0.5012 x^{1.9977}$

Q: The voltage V across a capacitor at time t seconds is given by the following table. Use the principle of least squares to fit a curve of the form $V = ae^{kt}$ to the data:

t	0	2	4	6	8
V	150	63	28	12	5.6

Ans: $V = 146.3 e^{-0.4118t}$